Yovanni Fabian Bernardino

Mechanical Engineering Student | yovannifabian8@gmail.com | (760) 390-4271

EDUCATION

University of California, Riverside | *B.S. Mechanical Engineering* | ***Expected Graduation: Spring 2028*** **Riverside, CA**

San Diego Miramar College | *A.S.: Pre-Engineering, Science, Chemistry, Physics* | ***Completed*** **San Diego, CA**

EXPERIENCE

Honda Materials Planning & Operations Student Associate - Spring 2027 **Upcoming**

Honda Auto Development Center Raymond, OH

Material R&D Intern **June - September 2025**

University of San Diego (USD) – NSF International Research Experience (IRES) Monterrey, MX & San Diego, CA

- Established a material-property database for a UV-curable photopolymer system by independently developing and evaluating 18+ resin formulations over 450+ hours of laboratory research, narrowing to three optimized formulations that characterize an ongoing DLP additive manufacturing research.
- Designed and executed an experimental characterization workflow by fabricating 12+ ASTM tensile specimens and performing tensile, DSC, TGA, and swelling analyses to quantify mechanical, thermal, and environmental performance, creating a repeatable methodology.
- Developed 20+ Python scripts to automate data processing, filtering, and visualization, generating 12+ engineering plots that identified formulation-property relationships to guide material optimization for a research thesis.
- Collaborated with an international team of five researchers as the only undergraduate participant, presenting findings at four international conferences in English and Spanish.

Data Analyst Intern **June - September 2024**

UC San Diego (UCSD) Scripps Institution of Oceanography – Whale Acoustics Lab San Diego, CA

- Developed a decision-support platform by synchronizing 8 months of continuous underwater acoustic recordings with thousands of AIS vessel records, enabling quantitative analysis of commercial vessel noise and its impact on marine ecosystems.
- Built and implemented an R Shiny application as a decision-support to automate vessel-noise data filtering, analysis, and visualization, reducing repetitive manual plotting workflows by ~90%.
- Integrated MATLAB analysis scripts to process large acoustic datasets, generating 20+ engineering visualizations that identified relationships between vessel speed, length, source level, and underwater radiated noise.

Material Development & Processing Intern **May 2024**

UC San Diego (UCSD) Materials Science and Engineering – Material Science Research Center San Diego, CA

- Selected as 1 of 10 participants from 800+ applicants for UCSD Materials Science & Engineering (SB-RIMSE), completing intensive semiconductor fabrication and characterization training.
- Fabricated porous silicon, photonic crystals, and silicon quantum dots through electrochemical anodization while following rigorous hydrofluoric acid (HF) safety protocols, laboratory procedures, and materials characterization techniques.
- Independently proposed and successfully synthesized a silicon quantum dot sample after reviewing scientific literature, demonstrating rapid learning, technical initiative, and experimental problem-solving.

Construction & Project Developer **June 2022 - January 2023**

Majestic Pools and Spas / Majestic Landscape San Diego, CA

- Constructed 40+ residential pools, 20+ drainage systems, gas utility installations, and retaining walls using engineering drawings, hand/power tools, laser measurement equipment, site layout, grading, and infrastructure installation.
- Designed residential pool and landscape layouts using Pool Studio, creating 3D visualizations and construction drawings that transform customer concepts into build-ready 3D models and construction documentation.

PERSONAL PROJECTS

Thermal Process Control & Data Acquisition System **Ongoing**

- Developing a Raspberry Pi-based thermal monitoring and control system using temperature probes and Python to automate data acquisition and improve temperature consistency during small-scale formulation processing.

Product Development – Water-Based Matte Pomade **Ongoing**

- Executed Design of Experiments (DOE) across 15+ water-based formulation trials, applying experimental techniques, thermal processing, and prototype testing to improve material performance and process consistency.

SKILLS

Engineering Methods: Design of Experiments (DOE), Design for Manufacturing (DFM), Process Development & Optimization, Data Visualization & Technical Writing, Cross-Functional Collaboration, Root Cause Analysis, Mechanical Design, Python, MATLAB, R, Excel, CAD